

Emilio Madrigal Mendoza

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EDUCATION

Duke University – Durham, NC

Bachelor of Science in Electrical and Computer Engineering

May 2028

RESEARCH EXPERIENCE:

MXene-Based Flying Robotics Enabled by Vapor Propulsion

Duke University – Durham, NC (January 2026 – Present)

- Fabricate $\text{Ti}_3\text{C}_2\text{T}_x$ MXene–agar composite films and develop laser-driven microrobotic propulsion systems.
- Design and test propulsion mechanisms, identifying asymmetric downward vapor jetting as the control method for rotational flight.
- Optimize material composition and NIR laser parameters to improve thrust efficiency and flight stability; characterize performance using optical and motion analysis instruments.

ENGINEERING PROJECTS:

ESP32 Open-Loop DC Motor Speed Controller — Embedded Control Systems

Personal Project

- Designed an open-loop PWM motor speed controller on ESP32 with potentiometer-based analog input, driving a 12V DC gear motor through an L298N H-bridge, and integrated an SSD1306 OLED for live telemetry (ADC value, PWM duty cycle, speed %, control source).
- Diagnosed and corrected ESP32 ADC non-linearity (effective ceiling ~2600/4095) by recalibrating the PWM mapping function and adding output clamping, eliminating control discontinuities at high duty cycle.
- Implemented a two-way Bluetooth Serial (SPP) interface for remote speed override via a custom command protocol, with a 3-second timeout failsafe that automatically reverts control to the potentiometer.

16-Bit RISC CPU – Processor Core Design

Duke University – Computer Architecture

- Designed a 16-bit single-cycle RISC CPU in Logisim with modular datapath and control logic (ALU, control unit, instruction decoder, register file), achieving full instruction-set coverage across arithmetic, logic, memory, and control operations
- Implemented Harvard architecture with separate instruction and data memory paths, eliminating data/instruction bus conflicts and validating correct execution of Fibonacci and Prime Generator programs in simulation.
- Integrated keyboard I/O and TTY output for interactive testing, reducing debug cycle time by enabling real-time assembly simulation feedback.

Blue Devil Grand Prix – Integrated Design

Duke University – Electric Circuits

- Designed and implemented an autonomous embedded system using Arduino, QTI sensors, and XBee communication for coordinated navigation.
- Led integration testing across a 5-person team, diagnosing 3 critical timing bugs in the XBee data-exchange protocol that had caused navigation failures in 2 of 5 prior test runs.

WORK EXPERIENCE:

Campus Mailbox Center Administrative Assistant

Duke University – Durham, NC (August 2024 – Present)

- Manage daily operations of the Campus Mailbox Center, tracking mail distribution via Excel and handling inquiries for 500+ students weekly.

VOLUNTEER EXPERIENCE:

Duke Life Peer Mentor

Duke University – Durham, NC (August 2025 – Present)

- Guide up to 5 first-year students per semester in academic planning and Duke resource navigation, with 100% mentee retention through the end of program cohort.

Academic Coach

Grow Your World – Duke, NC (Sep 2024 – Present)

- Deliver one-on-one STEM tutoring to underrepresented high school students, improving average client grades by one full letter grade over a single semester based on self-reported outcomes.

TECHNICAL SKILLS:

Hardware: Circuit Design, Power Electronics, Embedded Systems, Arduino, XBee

Tools: MATLAB/Simulink, Logisim, CAD

Languages: Python, C/C++, Assembly

Software: React Native, Supabase, Git, Linux, Microsoft Excel

Spoken Languages: Spanish, English.